

Measures of Shape (Detailed)

Measures of Shape are statistical measures that describe the **shape of a data distribution**. They tell us whether the data is **symmetric or skewed** and whether it has **light or heavy tails**.

In **Data Science**, Measures of Shape help us understand the distribution of data before applying statistical methods or machine learning algorithms.

Definition

Measures of Shape describe the overall form of a data distribution, including its symmetry, skewness, and peakedness.

They answer questions like:

- Is the data symmetric?
 - Is it skewed to the left or right?
 - Does it have many extreme values?
 - Is the distribution flat or sharply peaked?
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Why are Measures of Shape Important?

They help us to:

- Understand the distribution of data.
 - Detect skewed datasets.
 - Identify outliers.
 - Choose appropriate statistical methods.
 - Decide whether data transformation is needed.
 - Improve Machine Learning model performance.
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Types of Measures of Shape

There are **two main measures**:

1. **Skewness**
 2. **Kurtosis**
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1. Skewness

Definition

Skewness measures the **asymmetry** of a data distribution.

A perfectly symmetrical distribution has **zero skewness**.

Formula (Population)

$$\begin{aligned} &[\\ &\text{\text{Skewness}}= \\ &\frac{\frac{1}{N}\sum (x_i-\mu)^3}{\sigma^3} \\ &] \end{aligned}$$

Where:

- (x_i) = Observation
- (μ) = Mean
- (σ) = Standard Deviation
- (N) = Population size

In practice, software such as **Python**, **Pandas**, **NumPy**, **SciPy**, and **Excel** usually computes skewness directly.

Types of Skewness

There are **three types**.

A. Symmetrical Distribution

In a symmetrical distribution:

- Mean = Median = Mode
- Left side and right side are mirror images.
- Skewness = 0

Example

```
      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * *
      *
```

Characteristics

- Mean = Median = Mode
- No skewness
- Bell-shaped distribution

Examples

- IQ scores (approximately)
- Measurement errors
- Heights of adults (often approximately)

B. Positive Skewness (Right Skew)

The **tail extends toward the right**.

Most observations are small.

A few large values pull the mean to the right.

Relationship

Mode < Median < Mean

Visualization

```
*****
****
***
**
*
 \
  \
   \
```

Select the **right-skewed** option in the visualization to see how the long right tail causes the **mean to move farther right than the median and mode**.

Example

Monthly incomes

```
₹20,000
₹22,000
₹25,000
```

₹27,000
₹5,00,000

Most people earn around ₹20,000–₹30,000.

One very high salary pulls the average upward.

Characteristics

- Mean > Median > Mode
- Long right tail
- High-value outliers

Examples

- Income
- House prices
- Company profits

C. Negative Skewness (Left Skew)

The **tail extends toward the left**.

Most observations are high.

A few low values pull the mean to the left.

Relationship

Mean < Median < Mode

Visualization

```
 /  
 /  
 /  
*  
**  
***  
****  
*****
```

In the same visualization above, choose the **left-skewed** option to see how the long left tail causes the **mean to move left of the median and mode**.

Example

Easy examination

95
96
97
98
60

Most students score high.

One low score creates a left tail.

Characteristics

- Mean < Median < Mode
- Long left tail
- Low-value outliers

Examples

- Easy exam marks
 - Retirement age in some datasets
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Summary of Skewness

Type	Mean	Median	Mode	Tail Direction
Symmetric	Mean = Median = Mode	Equal	Equal	No tail
Positive Skew	Mean > Median > Mode	Middle	Lowest	Right
Negative Skew	Mean < Median < Mode	Middle	Highest	Left

Interpretation of Skewness Value

Skewness Value	Interpretation
0	Perfectly symmetric
-0.5 to 0.5	Approximately symmetric
0.5 to 1	Moderately positively skewed
Greater than 1	Highly positively skewed
-0.5 to -1	Moderately negatively skewed
Less than -1	Highly negatively skewed

Applications of Skewness

- Detecting outliers
- Checking data normality
- Feature engineering

- Financial risk analysis
 - Data transformation (Log, Box-Cox)
-

2. Kurtosis

Definition

Kurtosis measures the **tailedness** (or heaviness of tails) of a distribution.

It indicates whether the distribution has:

- More extreme values (heavy tails)
- Fewer extreme values (light tails)

It is **not simply a measure of how "peaked" the distribution is**—it primarily reflects the likelihood of extreme observations.

Formula

[

\text{Kurtosis}

$$\frac{1}{N} \sum (x_i - \mu)^4 / \sigma^4$$

]

Many statistical packages report **Excess Kurtosis**:

[

\text{Excess Kurtosis}

$$\text{Kurtosis} - 3$$

]

For a normal distribution:

- Kurtosis = **3**
 - Excess Kurtosis = **0**
-

Types of Kurtosis

A. Mesokurtic

Normal distribution.

- Kurtosis = 3
- Excess Kurtosis = 0

Characteristics

- Moderate tails
- Moderate number of outliers

Examples

- Many naturally occurring measurements
-

B. Leptokurtic

Higher kurtosis than normal.

Characteristics

- Heavy tails
- More outliers
- Extreme values occur more often

Examples

- Stock market returns
 - Financial data
-

C. Platykurtic

Lower kurtosis than normal.

Characteristics

- Light tails
- Fewer outliers
- Data spread more evenly

Examples

- Uniform distributions
-

Kurtosis Comparison

Type	Excess Kurtosis	Tails	Outliers

Leptokurtic	> 0	Heavy	Many
Mesokurtic	$= 0$	Normal	Moderate
Platykurtic	< 0	Light	Few

Difference Between Skewness and Kurtosis

Feature	Skewness	Kurtosis
Measures	Symmetry	Tail heaviness (extreme values)
Indicates	Left or right skew	Frequency of extreme values
Values	Negative, Zero, Positive	Often reported as excess kurtosis (negative, zero, positive)
Importance	Distribution shape	Outlier tendency

Real-Life Examples

Income Distribution

- Usually **positively skewed**.
 - A few people earn extremely high incomes.
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Stock Market Returns

Often **leptokurtic**.

Large gains or losses happen more often than a normal distribution predicts.

Student Marks

- Difficult exam → Positive skew.
 - Easy exam → Negative skew.
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Heights

Often approximately symmetric and close to mesokurtic.

Python Example

```
import pandas as pd

data = {
    "Marks": [45, 50, 55, 60, 65, 70, 75, 80, 95]
}

df = pd.DataFrame(data)

print("Skewness:", df["Marks"].skew())
print("Kurtosis:", df["Marks"].kurt())
```

Output (Example)

```
Skewness: 0.58
Kurtosis: -0.35
```

Interpretation:

- **Skewness = 0.58** → Moderately positively skewed.
 - **Kurtosis = -0.35** → Slightly platykurtic (lighter tails than a normal distribution).
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Applications in Data Science

Measures of Shape are used in:

- **Exploratory Data Analysis (EDA)**
 - Data preprocessing
 - Detecting outliers
 - Checking assumptions for statistical tests
 - Feature engineering
 - Selecting transformations (log, square root, Box-Cox)
 - Financial risk modeling
 - Machine Learning model preparation
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Summary

Measure	Meaning	Main Purpose
Skewness	Measures asymmetry of a distribution	Identify left or right skew
Kurtosis	Measures tail heaviness and likelihood of extreme values	Detect outlier-prone distributions

Key Takeaways

- **Measures of Shape** describe the **form** of a data distribution.
- **Skewness** tells whether the distribution is **symmetric, positively skewed, or negatively skewed**.
- **Kurtosis** indicates whether the distribution has **heavy, normal, or light tails**, which relates to the presence of extreme values.
- In **Data Science**, understanding skewness and kurtosis helps in choosing appropriate preprocessing techniques, identifying outliers, validating statistical assumptions, and building more reliable machine learning models.

Link : [CHATGPT](#)